THE CHAIN RULE

$y' = f'(g(x)) \cdot g'(x)$

Note: The composite function $(f o g)$ is defined by $(f o g)(x) = f(g(x))$

Example 1: Suppose $f(x) = \sqrt{x}$ and $g(x) = x - 2$

a) Express $(f o g)(x)$

$(f \circ g)(x) = \sqrt{x-2}$

b) Evaluate $f(g(4))$

$f(g(4)) = \sqrt{2}$

c) Express $(g o f)(x)$

$(g \circ f)(x) = \sqrt{x} - 2$

d) Evaluate $g(f(4))$

$g(f(4)) = 0$

Example 2: Differentiate the function $h(x) = (2x^3 + 5x)^{35}$.

$h'(x) = 35(2x^3+5x)^{34} \cdot (6x^2+5)$

Note:

If we let $f(u) = u^{35}$ and we let $u = g(x) = 2x^3 + 5x$ then $h(x) = f(g(x))$.

Since we know how to differentiate $f(u)$ and $g(x)$ individually, the **chain rule** allows us to differentiate $y = f(g(x))$.

It turns out that the derivative of the composite function, $h(x) = f(g(x))$ is the product of the derivatives of $f(u)$ and $g(x)$.

That is, $h'(x) = f'(u)g'(x)$.

$h'(x) = 35u^{34} \cdot (6x^2+5)$

$h'(x) = 35(2x^3+5x)^{34} \cdot (6x^2+5)$

Recall:

$\frac{dy}{du}$ is the rate of change of y with respect to u.

$\frac{du}{dx}$ is the rate of change of u with respect to x.

$\frac{dy}{dx}$ is the rate of change of y with respect to x.

If $g(x)$ is differentiable at x and $f(x)$ is differentiable at $g(x)$, then the composite function, $h(x)=f(g(x))$ or $h(x)=(fog)(x)$ is differentiable at x and $h'(x)$ is given by the product $h'(x)=f'(g(x))g'(x)$.

In Leibniz notation, If $y=f(u)$ and $u=g(x)$, are both differentiable functions, then

$$\frac{dy}{dx}=\frac{dy}{du}\times\frac{du}{dx}.$$

Examples:

1. If $y=2u^2+6u$, $u=2+v^3$, and $v=\sqrt{x}$ then evaluate $\frac{dy}{dx}$ at $x=4$.

$$\frac{dy}{dx}=\frac{dy}{du}\cdot\frac{du}{dv}\cdot\frac{dv}{dx}$$

when $x=4$,

$v=2$,

$u=10$

$$\frac{dy}{dx}=(4u+6)\cdot(3v^2)\cdot\frac{1}{2\sqrt{x}}$$

$$\therefore \left.\frac{dy}{dx}\right|_{x=4}=\frac{(46)(12)}{4}\cdot=138$$

2. Determine the derivative of $f(x)=\sqrt{\frac{x-3}{3-5x}}$

let $u=g(x)=\frac{x-3}{3-5x}$

$\therefore h(u)=\sqrt{u}$

$$* \quad f'(x)=h'(u)\cdot g'(x)$$

$$=\frac{1}{2\sqrt{u}}\cdot\left[\frac{(1)(3-5x)+5(x-3)}{(3-5x)^2}\right]$$

$$=\frac{-12}{2\sqrt{\frac{x-3}{3-5x}}\cdot(3-5x)^2}$$

$$=\frac{-6}{\sqrt{x-3}\cdot\sqrt{(3-5x)}^3}$$

$$=\frac{-6}{\sqrt{(x-3)(3-5x)^3}}$$

3. Differentiate:

a) $f(x)=\left(4x^2+3x\right)^5$

$f'(x) = 5\left(4x^2+3x\right)^4 \cdot (8x+3)$

b) $f(x)=\left(5x^2-x\right)^\pi$

$f'(x) = \pi\left(5x^2-x\right)^{\pi-1} \cdot (10x-1)$

c) $y=\dfrac{x}{1-4x^2}$

$$y' = \frac{1-4x^2 - x(-8x)}{(1-4x^2)^2}$$

$$y' = \frac{1+4x^2}{(1-4x^2)^2}$$

d) $f(t)=(2t^3+6t-5)^5$

$f'(t) = 5\left(2t^3+6t-5\right)^4\left(6t^2+6\right)$

$f'(t) = 30\left(2t^3+6t-5\right)^4\left(t^2+1\right)$

e) $y=(3x^2-5)\sqrt{2x^2+9x}$

$\left(2x^2+9x\right)^{\frac{1}{2}}$

$$\frac{dy}{dx} = 6x\sqrt{2x^2+9x} + \frac{(3x^2-5)\cdot(4x+9)}{2\sqrt{2x^2+9x}}$$

$$\frac{dy}{dx} = \frac{12x(2x^2+9x) + (3x^2-5)(4x+9)}{2\sqrt{2x^2+9x}}$$

$$\frac{dy}{dx} = \frac{24x^3+108x^2+12x^3+27x^2-20x-45}{2\sqrt{2x^2+9x}}$$

$$\frac{dy}{dx} = \frac{36x^3+135x^2-20x-45}{2\sqrt{2x^2+9x}}$$

f) $m(t)=\sqrt[3]{1+t+t^2}$

$\left(1+t+t^2\right)^{\frac{1}{3}}$

$$m'(t) = \frac{1}{3\cdot\sqrt[3]{(1+t+t^2)^2}}\cdot(1+2t)$$

$$= \frac{2t+1}{3\cdot\sqrt[3]{(1+t+t^2)^2}}$$

4. a) If $h(x)=\dfrac{(f(x))^2}{g(x)}$, determine $h'(x)$.

$$h'(x) = \frac{2f(x)\cdot f'(x)\cdot g(x) - (f(x))^2\cdot g'(x)}{[g(x)]^2}$$

b) Given $f(1)=2$, $f'(1)=-3$, $g(1)=1$ and $g'(1)=4$, determine $h'(1)$.

$$h'(1) = \frac{2f(1)\cdot f'(1)g(1) - (f(1))^2\cdot g'(1)}{[g(1)]^2}$$

$$h'(1) = \frac{2(2)(-3)(1) - (2)^2\cdot(4)}{(1)^2}$$

$$h'(1) = -12 - 16$$

$$h'(1) = -28$$